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GATE ECE · ELECTRONIC DEVICES AND VLSI · CHAPTER-WISE PYQS

PN Junction

GATE ECE Electronic Devices and VLSI · Previous Year Questions with Answer Key

42	27	1987 – 2026	English
QUESTIONS	PAPERS COVERED	YEARS	LANGUAGE

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Marks 1	18 questions
Marks 2	24 questions
Answer key	at the end

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Marks 1 18 questions

Q1. GATE ECE 2026

Consider carrier transport in a Zener diode in the breakdown region. Which is the dominant transport mechanism for current flow in this case?

(A) Drift
(B) Diffusion
(C) Tunneling
(D) Ballistic transport

Open solution

Q2. GATE ECE 2026

Consider a p-n junction diode when it is forward biased with 2 V . Which of the following is/are the correct magnitude(s) of the energy difference between quasi Fermi-levels, E_{fn} in the n -side and E_{fp} in the p-side?

(A) 2 eV (B) 1 eV (C) 2 V (D) 1 V

Open solution

Q3. GATE ECE 2021

A silicon $P-N$ junction is shown in the figure. The doping in the P region is $5 \times 10^{16}\ \mathrm{cm}^3$ and doping in the N region is $10 \times 10^{-16}\ \mathrm{cm}^{-3}$. The parameters given are

Built-in voltage $(\phi_{bi}) = 0.8\ \mathrm{V}$

Electro charge $(q) = 1.6 \times 10^{-19}\mathrm{C}$

Vacuum permittivity of silicon $(\varepsilon_{si}) = 12$



The magnitude of reverse bias voltage that would completely deplete one of the two regions (P or N) prior to the other (rounded off to one decimal place) is ____ V.

Open solution

Q4. GATE ECE 2020

Consider the recombination process via bulk traps in a forward biased pn homojunction diode. The maximum recombination rate is $U_{\max}$. If the electron and the hole capture cross sections are equal, which one of the following is FALSE?

(A) $U_{\max}$ occurs at the edges of the depletion region in the device.
(B) $U_{\max}$ depends exponentially on the applied bias.
(C) With all other parameters unchanged, $U_{\max}$ increases if the thermal velocity of carrier increases.
(D) With all other parameters unchanged, $U_{\max}$ decreases if the intrinsic carrier density is reduced.

Open solution

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Q5. GATE ECE 2015 Set 1

A region of negative differential resistance is observed in the current voltage characteristics of a silicon PN junction if

(A) Both the P-region and the N-region are heavily doped

(B) The N-region is heavily doped compared to the P-region

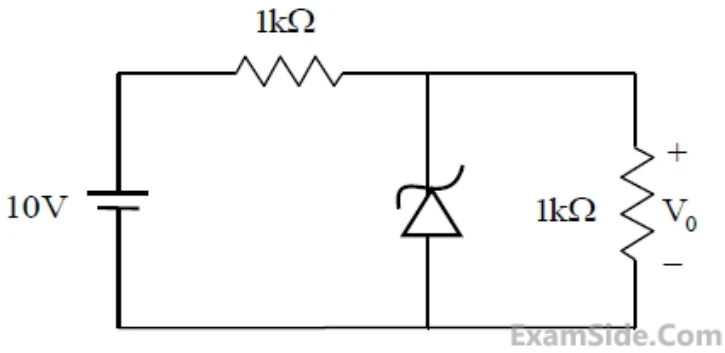
(C) The P-region is heavily doped compared to the N-region

(D) An intrinsic silicon region is inserted between the P-region and the N-region

Open solution

Q6. GATE ECE 2015 Set 1

In the circuit shown below, the Zener diode is ideal and the Zener voltage is 6V. The output voltage V_0 (in volts) is _____.



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Open solution

Q7. GATE ECE 2014 Set 1

When the optical power incident on a photodiode is 10μW and the responsivity is 0.8 A/W, the photocurrent generated (in μA) is _____.

Open solution

Q8. GATE ECE 2013

In a forward biased P-N junction diode, the sequence of events that best describes the mechanism of current flow is

(A) injection, and subsequent diffusion and recombination of minority carriers

(B) injection, and subsequent drift and generation of minority carriers

(C) extraction, and subsequent diffusion and generation of minority carriers

(D) extraction, and subsequent drift and recombination of minority carriers

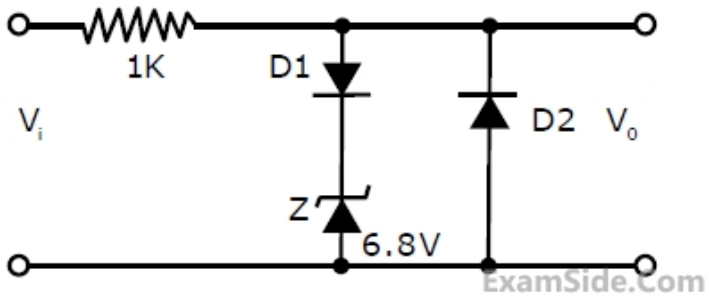
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Q9. GATE ECE 2008

In the following limiter circuit, an input voltage $V_i = 10\sin(100\pi t)$ applied. Assume that the diode drop is 0.7V when it is forward biased. The Zener breakdown voltage is 6.8V.

Open solution



The maximum and minimum values of the output voltage respectively are

(A) 6.1V, – 0.7V (B) 0.7V, – 7.5V (C) 7.5V, – 0.7V (D) 7.5V, – 7.5V

Q10. GATE ECE 2008

Which of the following is NOT associated with a P-N junction?

(A) Junction capacitance
(B) Charge Storage Capacitance
(C) Depletion Capacitance
(D) Channel Length Modulation

Open solution

Q11. GATE ECE 2007

In a p^+n junction diode under reverse bias, the magnitude of electric field is maximum at

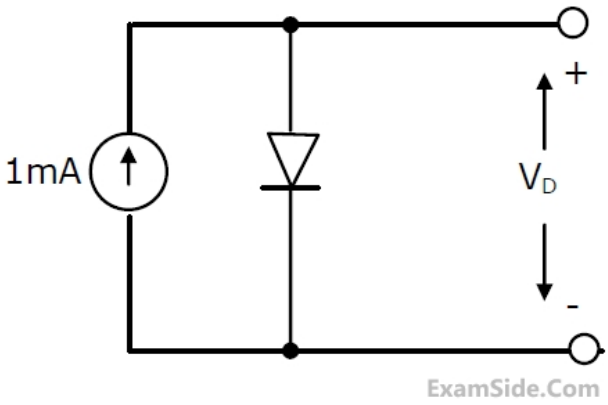
(A) the edge of the depletion region on the p-side
(B) the edge of the depletion region on the n-side
(C) the p^+n junction
(D) the center of the depletion region on the n-side

Open solution

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Q12. GATE ECE 2002

In the figure, a silicon diode is carrying a constant current of 1 mA. When the temperature of the diode is 20°C, V_D is found to be 700 mV. If the temperature rises to 40°C, V_D becomes approximately equal to



Open solution

(A) 740 mV (B) 660 mV (C) 680 mV (D) 700 mV

Q13. GATE ECE 1998

For small signal a.c. operation, a practical forward biased diode can be modeled as

(A) a resistance and a capacitance in series

(B) an ideal diode and resistance in parallel

(C) a resistance and an ideal diode in series

(D) a resistance

Open solution

Q14. GATE ECE 1998

The static characteristic of an adequately forward biased p-n junction is a straight line, if the plot is of

(A) log I vs. log V (B) log I vs. V (C) I vs. log V (D) I vs. V

Open solution

Q15. GATE ECE 1995

The diffusion potential across a P-N junction

(A) decreases with increasing doping concentration

(B) increases with decreasing band gap

(C) does not depend on doping concentration

(D) increases with increase in doping concentrations

Open solution

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Q16. GATE ECE 1995

A zener diode works on the principle of

(A) tunneling of charge carriers across the junction

(B) thermo ionic emission

(C) diffusion of charge carriers across the junction

(D) hopping of charge carriers across the junction

Open solution

Q17. GATE ECE 1995

The depletion capacitance, C_J, of an abrupt P-N junction with constant doping on either side varies with reverse bias, V_R , as

(A) $C_J \propto V_R$ (B) $C_J \propto V_R^{-1}$ (C) $C_J \propto V_R^{-1/2}$ (D) $C_J \propto V_R^{-1/3}$

Open solution

Q18. GATE ECE 1990

In a junction diode

(A) The depletion capacitance increases with increase in the reverse bias

(B) The depletion capacitance decreases with increase in the reverse bias

(C) The depletion capacitance increases with increase in the forward bias

(D) The depletion capacitance is much higher than the depletion capacitance when it is forward bias.

Open solution

Marks 2 24 questions

Q19. GATE ECE 2026

Consider an LED based on a direct bandgap semiconductor material with energy bandgap 1.3 eV . Given: Plank's constant, $h = 6.63 \times 10^{-34}\ \mathrm{J\,s}$ and speed of light in free space is $3 \times 10^{8}\ \mathrm{m\,s^{-1}}$.

In which of the following wavelength ranges the LED will NOT emit?

(A) $1410 \pm 20\ \mathrm{nm}$ (B) $1090 \pm 20\ \mathrm{nm}$ (C) $950 \pm 20\ \mathrm{nm}$ (D) $510 \pm 20\ \mathrm{nm}$

Open solution

Q20. GATE ECE 2025

An ideal p-n junction germanium diode has a reverse saturation current of $10\mu\ \mathrm{A}$ at 300 K . The voltage (in Volts, rounded off to two decimal places) to be applied across the junction to get a forward bias current of 100 mA at 300 K is _____. (Consider the Boltzmann constant $k_B = 1.38 \times 10^{-23}\ \mathrm{J/K}$ and the charge of an electron $e = 1.6 \times 10^{-19}\mathrm{C}$.)

Open solution

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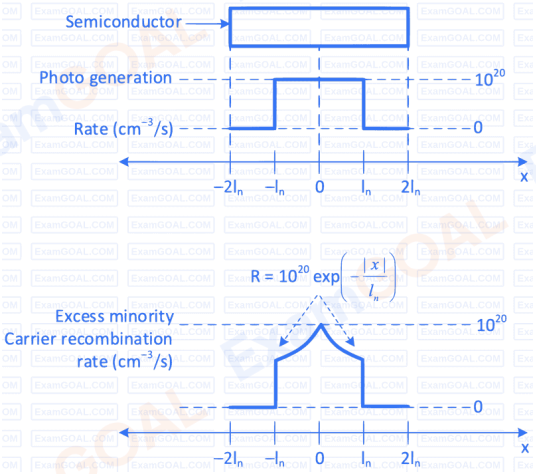
Q21. GATE ECE 2024

The photocurrent of a PN junction diode solar cell is 1 mA. The voltage corresponding to its maximum power point is 0.3 V. If the thermal voltage is 30 mV, the reverse saturation current of the diode (in nA, *rounded off to two decimal places*) is _____.

Open solution

Q22. GATE ECE 2022

A p-type semiconductor with zero electric field is under illumination (low level injection) in steady state condition. Excess minority carrier density is zero at x = $\pm 2l_n$, where $l_n = 10^{-4}$ cm is the diffusion length of electrons. Assume electronic charge, q = -1.6×10^{-19} C. The profiles of photo-generation rate of carriers and the recombination rate of excess minority carriers (R) are shown. Under these conditions, the magnitude of the current density due to the photo-generated electrons at x = $+2l_n$ is _____ mA/cm^2 (rounded off to two decimal places).



Open solution

Q23. GATE ECE 2020

A pn junction solar cell of area $1.0\ \mathrm{cm}^2$, illuminated uniformly with $100\mathrm{mWcm}^{-2}$; has the following parameter : Efficiency $= 15\%$, open circuit voltage $= 0.7$ V, fill factor $= 0.8$, and thickness $= 200\mu$ m. The charge of an electron is 1.6×10^{-19}C. The average optical generation rate ($\mathrm{incm}^{-3}\,\mathrm{s}^{-1}$) is

(A) 0.84×10^{19} (B) 83.60×10^{19} (C) 1.04×10^{19} (D) 5.57×10^{19}

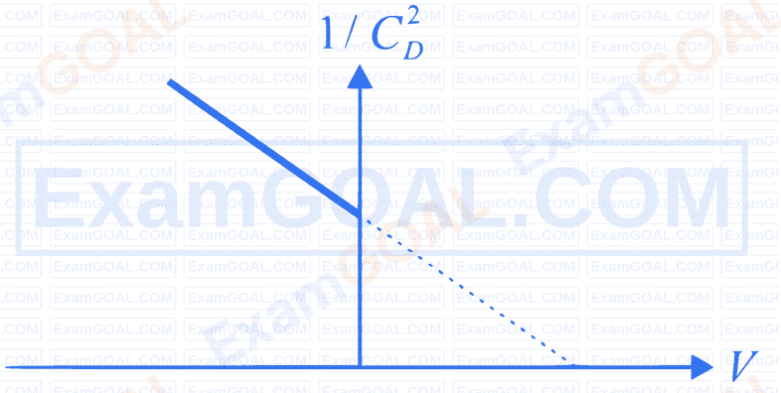
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Q24. GATE ECE 2020

Open solution

A one-sided abrupt pn junction diode has a depletion capacitance C_D of 50 pF at a reverse bias 0.2 V . The plot of $\frac{1}{C_D^2}$ versus the applied voltage V for this diode is a straight line as shown in the figure below. The slope of the plot is ____ $\times 10^{20}\ \mathrm{F}^{-2}\ \mathrm{V}^{-1}$.



(A) -1.2 (B) -5.7 (C) -0.4 (D) -3.8

Q25. GATE ECE 2017 Set 2

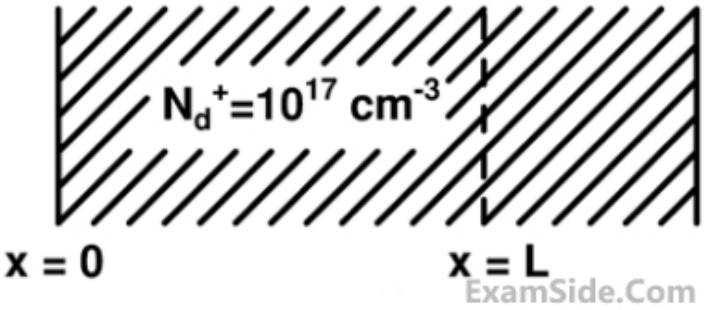
Open solution

For a particular intensity of incident light on a silicon pn junction solar cell, the photocurrent density (J_L) is 2.5 mA/cm^2 and the open-circuit voltage (V_{oc}) is 0.451 V. consider thermal voltage (V_T) to be 25 mV. If the intensity of the incident light is increased by 20 times, assuming that the temperature remains unchanged. V_{oc} (in volts) will be _____.

Q26. GATE ECE 2016 Set 2

Open solution

Consider a region of silicon devoid of electrons and holes, with an ionized donor density of $N_d^+ = 10^{17}\ \mathrm{cm}^{-3}$. The electric field at x = 0 is 0 V/cm and the electric field at x = L is 50 kV/cm in the positive x direction. Assume that the electric field is zero in the y and z directions at all points.



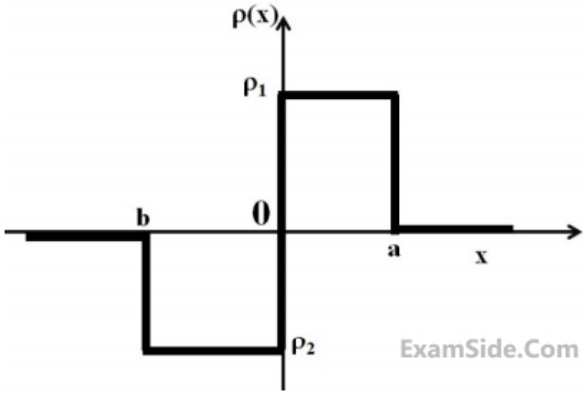
Given q = 1.6 × 10^{-19} coulomb, ε_0 = 8.85 × 10^{-14} F/cm, ε_r = 11.7 for silicon, the value of L in nm is _____.

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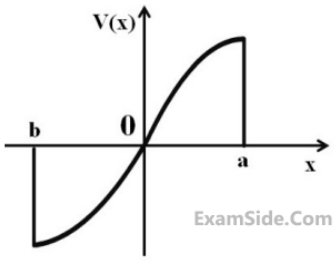
Q27. GATE ECE 2016 Set 3

Open solution

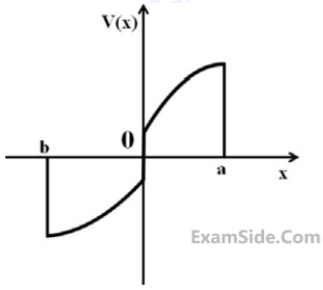
Consider the charge profile shown in the figure. The resultant potential distribution is best described by



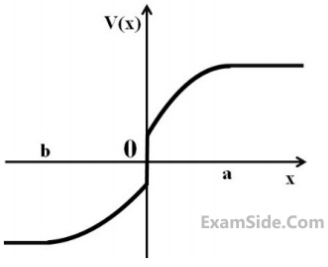
(A)



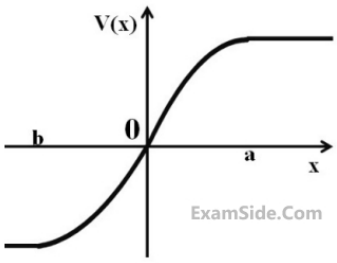
(B)



(C)



(D)

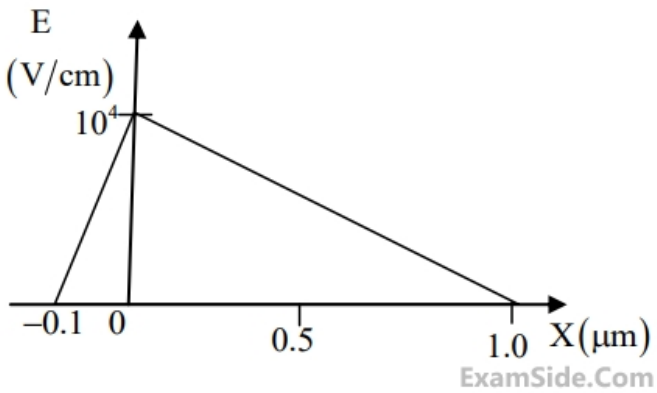


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Q28. GATE ECE 2015 Set 3

Open solution

The electric field profile in the depletion region of a p-n junction in equilibrium is shown in the figure. Which one of the following statements is NOT TRUE?



(A) The left side of the junction is n-type and the right side is p-type

(B) Both the n-type and p-type depletion regions are uniformly doped

(C) The potential difference across the depletion region is 700 mV

(D) If the p-type region has a doping concentration of 10^{15} cm^{-3}, then the doping concentration in the n-type region will be 10^{16} cm^{-3}

Q29. GATE ECE 2015 Set 1

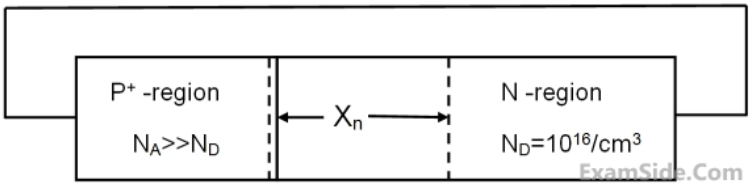
Open solution

The built-in potential of an abrupt p-n junction is 0.75V. If its junction capacitance (C_J) at a reverse bias (V_R) of 1.25V is 5pF, the value of C_J (in pF) when V_R = 7.25V is _____.

Q30. GATE ECE 2014 Set 2

Open solution

Consider an abrupt PN junction (at T = 300 K) shown in the figure. The depletion region width X_n on the N-side of the junction is 0.2 μm and the permittivity of silicon (ε_{si}) is 1.044×10^{-12} F/cm. At the junction, the approximate value of the peak electric field (in kV/cm) is _____.

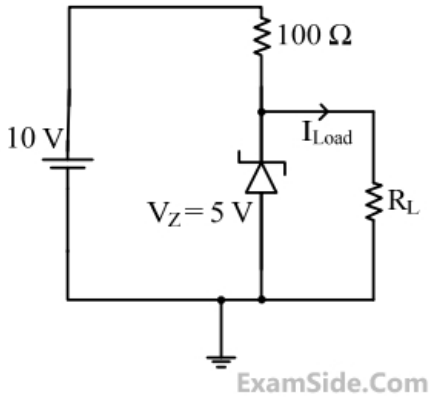


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Q31. GATE ECE 2013

Open solution

In the circuit shown below, the knee current of the ideal Zener diode is 10 mA. To maintain 5V across R_L, the minimum value of R_L in Ω and the minimum power rating of the Zener diode in mW, respectively, are

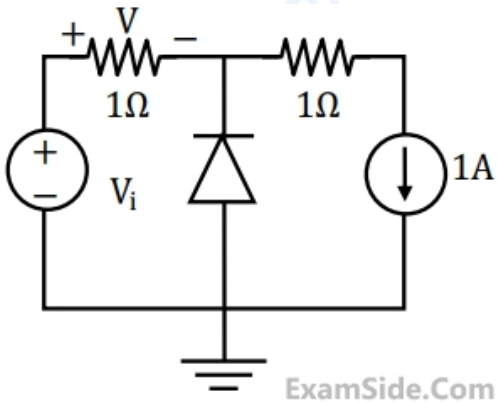


(A) 125 and 125 (B) 125 and 250 (C) 250 and 125 (D) 250 and 250

Q32. GATE ECE 2009

Open solution

In the circuit below, the diode is ideal. The voltage V is given by



(A) Min (V_i, 1) (B) Max (V_i, 1) (C) Min (-V_i, 1) (D) Max (-V_i, 1)

Q33. GATE ECE 2007

Open solution

A P^+-N junction has a built-in potential of 0.8 V. The depletion layer width at a reverse bias of 1.2V is 2 μm. For a reverse bias of 7.2 V, the depletion layer width will be:

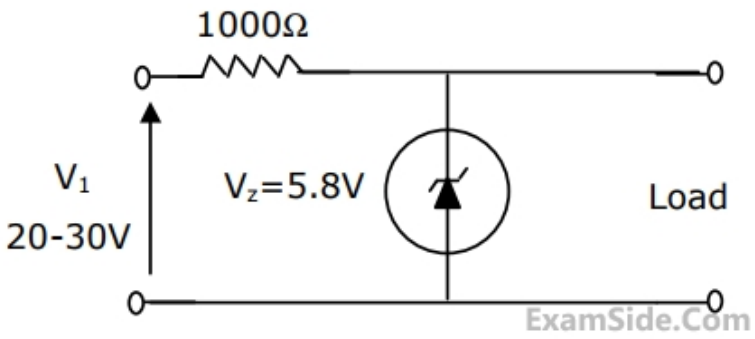
(A) 4 μm (B) 4.9 μm (C) 8 μm (D) 12 μm

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Q34. GATE ECE 2005

The Zener diode in the regulator circuit shown in figure has a Zener voltage of 5.8 Volts and a Zener knee current of 0.5 mA. The maximum load current drawn from this circuit ensuring proper functioning over the input voltage range between 20 and 30 Volts, is

Open solution



(A) 23.7 mA (B) 14.2 mA (C) 13.7 mA (D) 24.2 mA

Q35. GATE ECE 2005

A Silicon PN junction diode under reverse bias has depletion region of width 10 μm. The relative permittivity of Silicon, ε_r = 11.7 and the permittivity of free space $\varepsilon_0 = 8.854 \times 10^{-12}$ F/m.The depletion capacitance of the diode per square meter is

Open solution

(A) 100 μF (B) 10 μF (C) 1 μF (D) 20 μF

Q36. GATE ECE 2003

At 300 K, for a diode current of 1 mA, a certain germanium diode requires a forward bias of 0.1435 V. Where as a certain silicon diode requires a forward bias of 0.718 V. Under the conditions stated above, the closest approximation of the ratio of reverse saturation current in germanium diode to that in silicon diode is

Open solution

(A) 1 (B) 5 (C) 4×10^3 (D) 8×10^3

Q37. GATE ECE 2003

Choose proper substitutes for X and Y to make the following statement correct Tunnel diode and Avalanche photodiode are operated in X bias and Y bias respectively.

Open solution

(A) X: reverse, Y: reverse (B) X: reverse, Y: forward
(C) X: forward, Y: reverse (D) X: forward, Y: forward

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Q38. GATE ECE 1993

The built-in potential (Diffusion Potential) in a p-n junction

(A) is equal to the difference in the Fermi level of the two sides, expressed in volts.

(B) increases with the increase in the doping levels of the two sides.

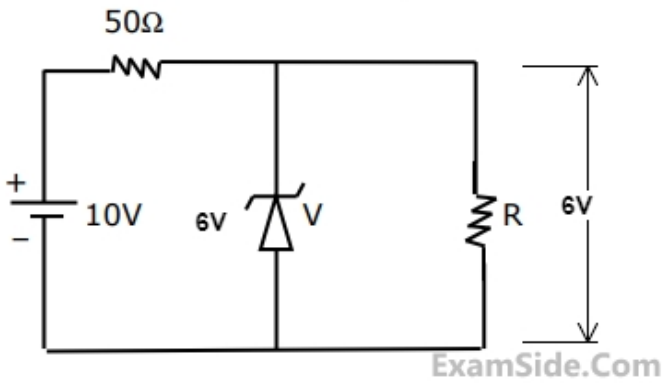
(C) increases with the increase in temperature.

(D) is equal to the average of the Fermi levels of the two sides.

Open solution

Q39. GATE ECE 1992

The 6 V Zener diode shown in figure has zero Zener resistance and a knee current of 5 mA. The minimum value of R, so that the voltage across it does not fall below 6 V is:



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(A) 1.2 kΩ (B) 50 Ω (C) 80 Ω (D) 0 Ω

Open solution

Q40. GATE ECE 1991

The small signal capacitances of an abrupt P^+–N junction is 1 nF at zero bias. If the built-in voltage is 1 volt, the capacitance at a reverse bias voltage of 99 volts is equal to

(A) 100 (B) 0.1 (C) 0.01 (D) 10

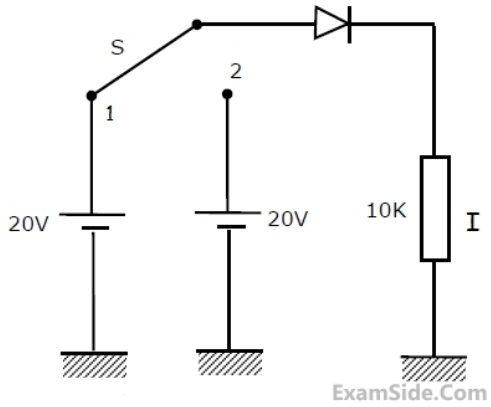
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Q41. GATE ECE 1991

Open solution

Referring to the figure. The switch S is in position 1 initially and steady state conditions exist from time t = 0 to t = t_0. At t = t_0, the switch is suddenly thrown into position 2. The current I through the 10K resistor as a function of time t, from t = 0 is (in mA) _____.
(Give the sketch showing the magnitudes of the current at t = 0, t = t_0 and t = ∞)



Q42. GATE ECE 1987

Open solution

The diffusion capacitance of a P-N junction

(A) Decreases with increasing current and increasing temperature

(B) Decreases with decreasing current and increasing temperature

(C) Increases with increasing current and increasing temperature

(D) Does not depend on current and temperature

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Answer Key

Marks 1

1. **C**	2. **A**	3. **8.8 to 9.0**	4. **A**	5. **A**
6. **5**	7. **8**	8. **A**	9. **C**	10. **D**
11. **C**	12. **B**	13. **D**	14. **B**	15. **D**
16. **A**	17. **C**	18. **B**		

Marks 2

19. **A, B, D**	20. **0.23 to 0.24**	21. **4.0 to 4.26**	22. **0.59**	23. **A**
24. **B**	25. **0.526**	26. **32.36**	27. **D**	28. **C**
29. **2.5**	30. **30.65**	31. **B**	32. **D**	33. **A**
34. **C**	35. **B**	36. **C**	37. **C**	38. **A, B**
39. **C**	40. **B**	41. **-2**	42. **B**	

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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